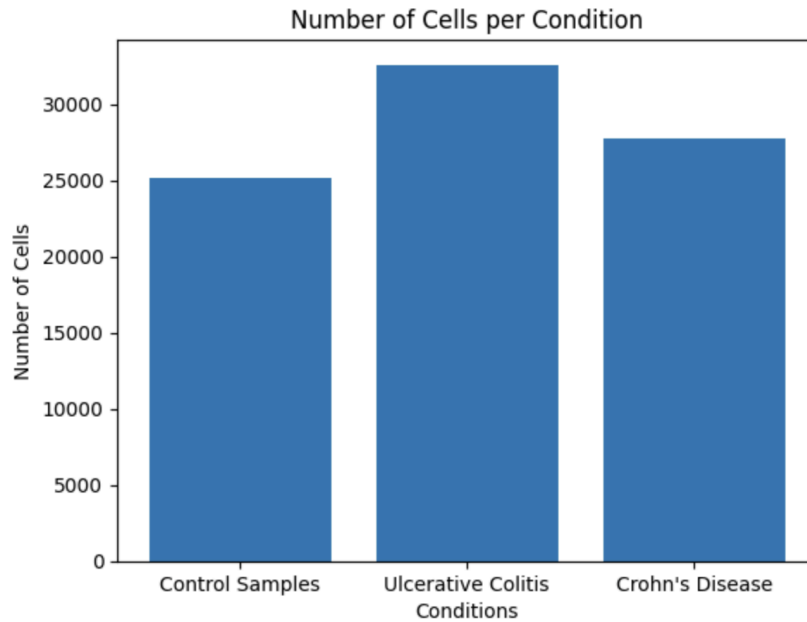
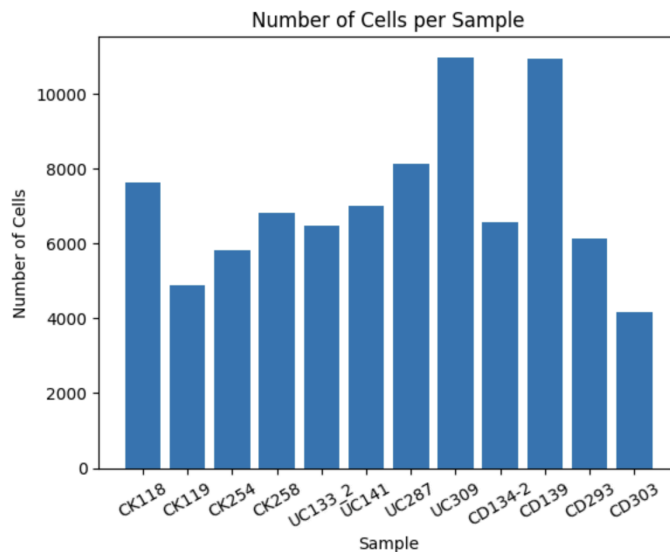


Figure 1: (Alex)



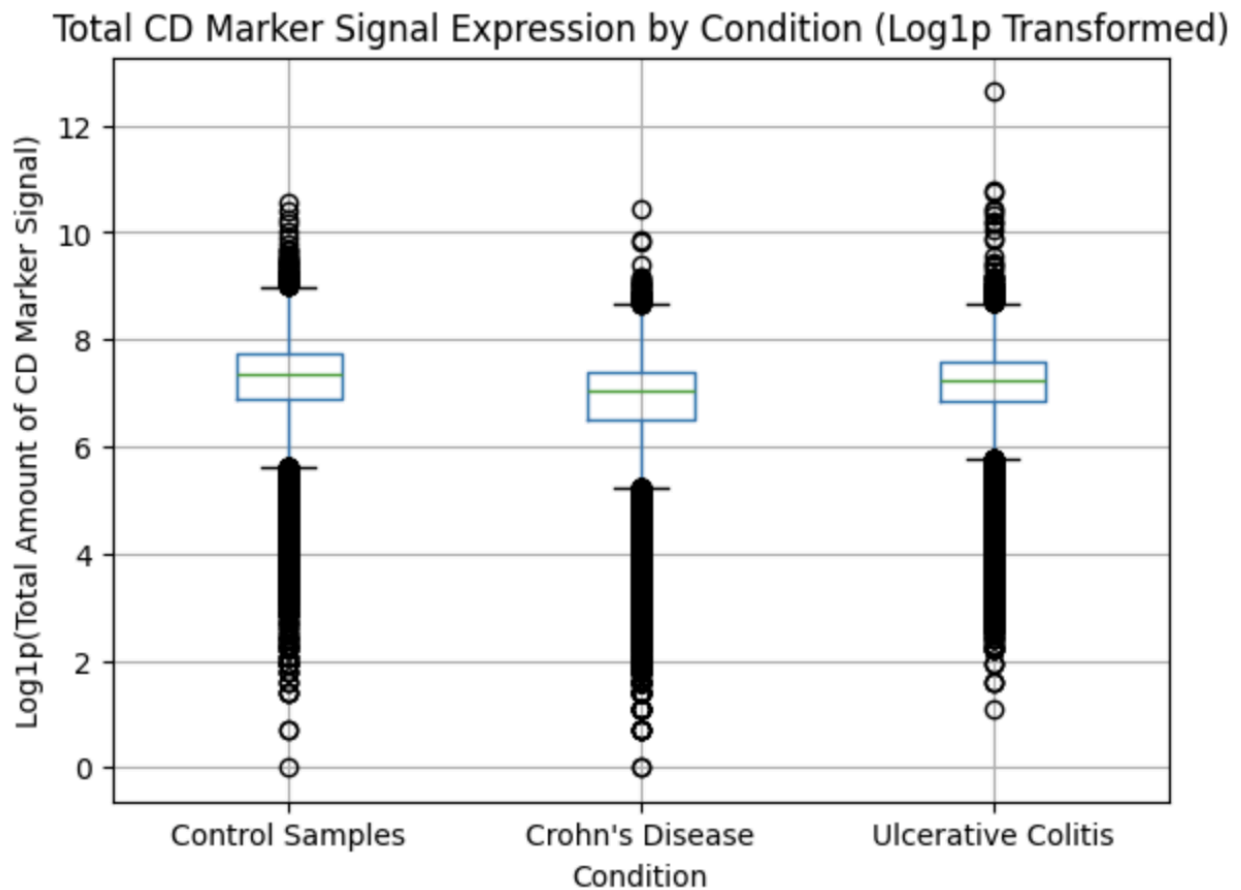
Description: For all samples, I added the different types of cells. From the graph, it is clear that there is not an equal amount of cells per condition. Ulcerative Colitis cells were the most common sampled, then Crohn's disease cells, and then control cells (healthy cells). No finding, this is just a broad summary of our data.

Figure 2: (Alex)



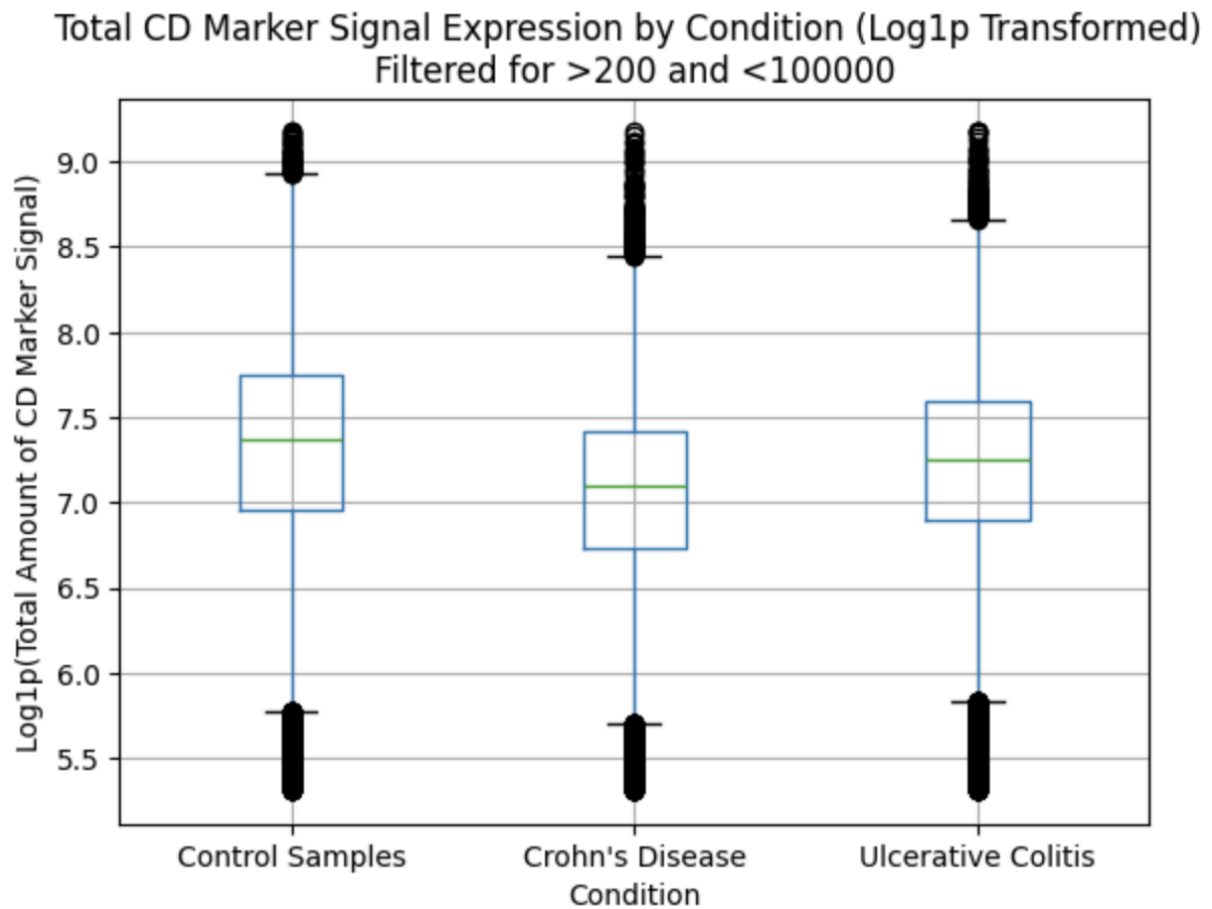
Description: There were 12 samples, for each sample I counted the amount of cells in the sample. This shows that not all samples were the same, some tested many cells, some tested a few.

Figure 3: (Alex)



Description: For all samples, I counted the average amount of CD marker signal for each of the conditions. This graph shows that the data is not that spread out, with each of the conditions having around the same amount of CD marker signal. However, there are still many outliers, both above and below, for all of the conditions as well.

Figure 4: (Alex)



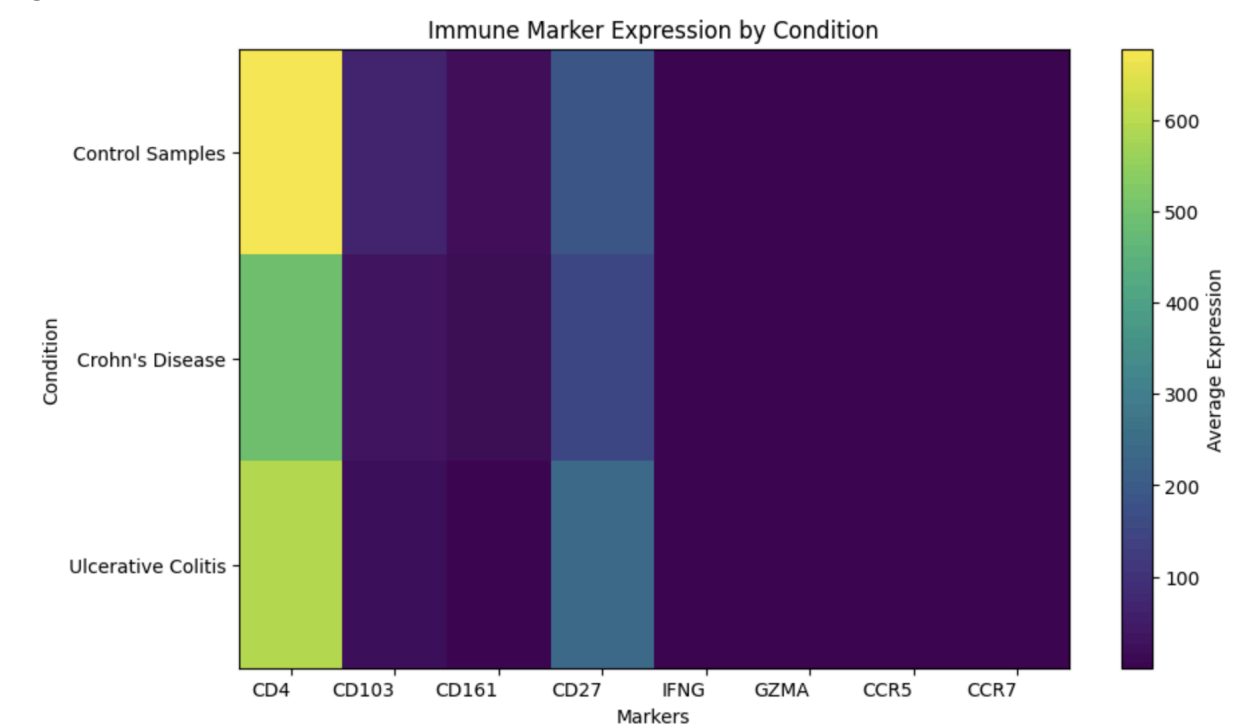
Description: This is the same as Figure 3, but filtered for samples >200 and <100000. When filtered, we can get a better look at the data and see that the medians are around the same, even when cutting out most of the lower outliers. This shows that the data is strongly centered around the median.

Figure 5: (Alex)



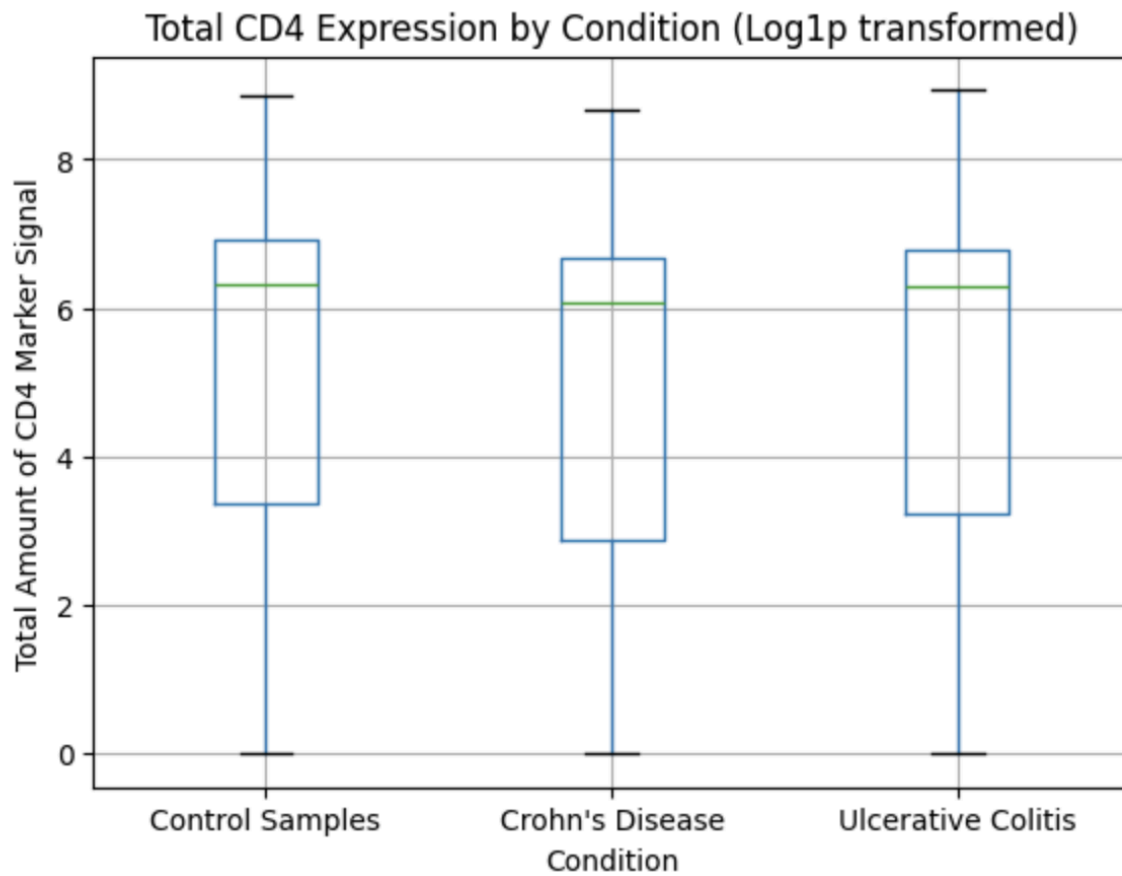
Description: At first I did a heatmap of all genes, but this heatmap is filtered for the genes with average expression higher than 50. The ones with the greatest expression are CD27, CD3, CD4, and CD8. However, only CD27 are markers. This heatmap shows that perhaps the markers are not the best indicators for the disease, but also calls for a further dive into the difference in expression for the higher expressed genes by condition.

Figure 6: (Alex)



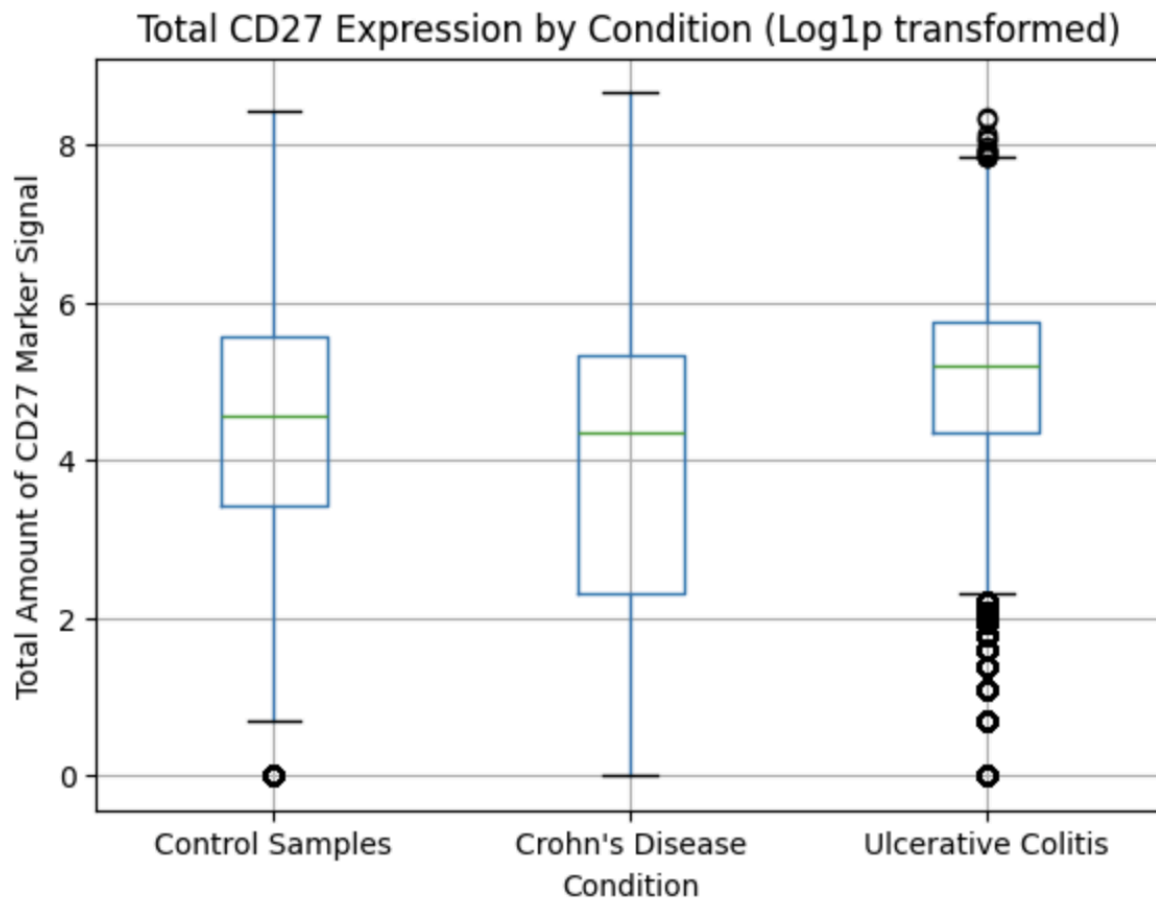
Description: This is a heatmap of marker expression by condition. This heatmap only shows the markers, whereas figure 6 showed all genes >50. There is very little difference in expression for many of the genes per conditions, with the only obvious difference in markers CD4 and CD27.

Figure 7: (Alex)



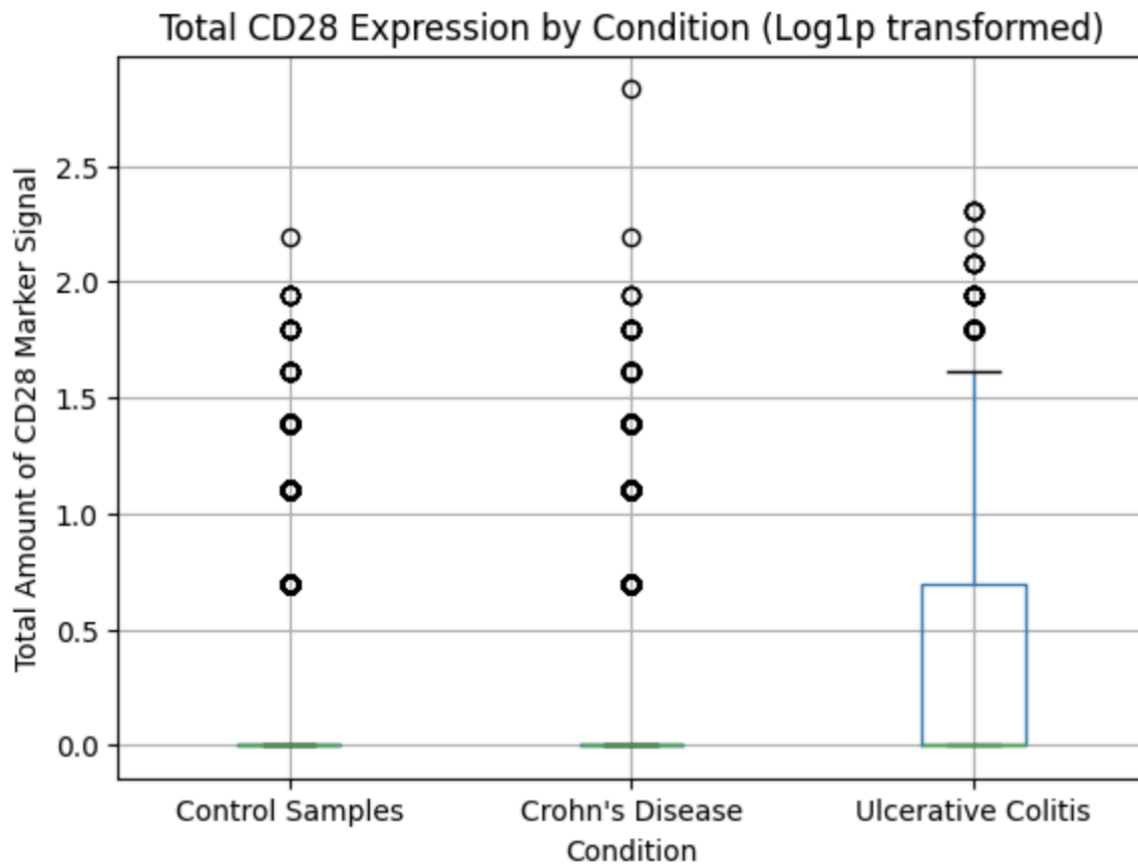
Description: I decided to explore the difference in expression in the CD4 gene. This shows the total CD4 expression by condition (log transformed). The box plots are quite similar for all 3 conditions, with no clear outliers or differences.

Figure 8: (Alex)



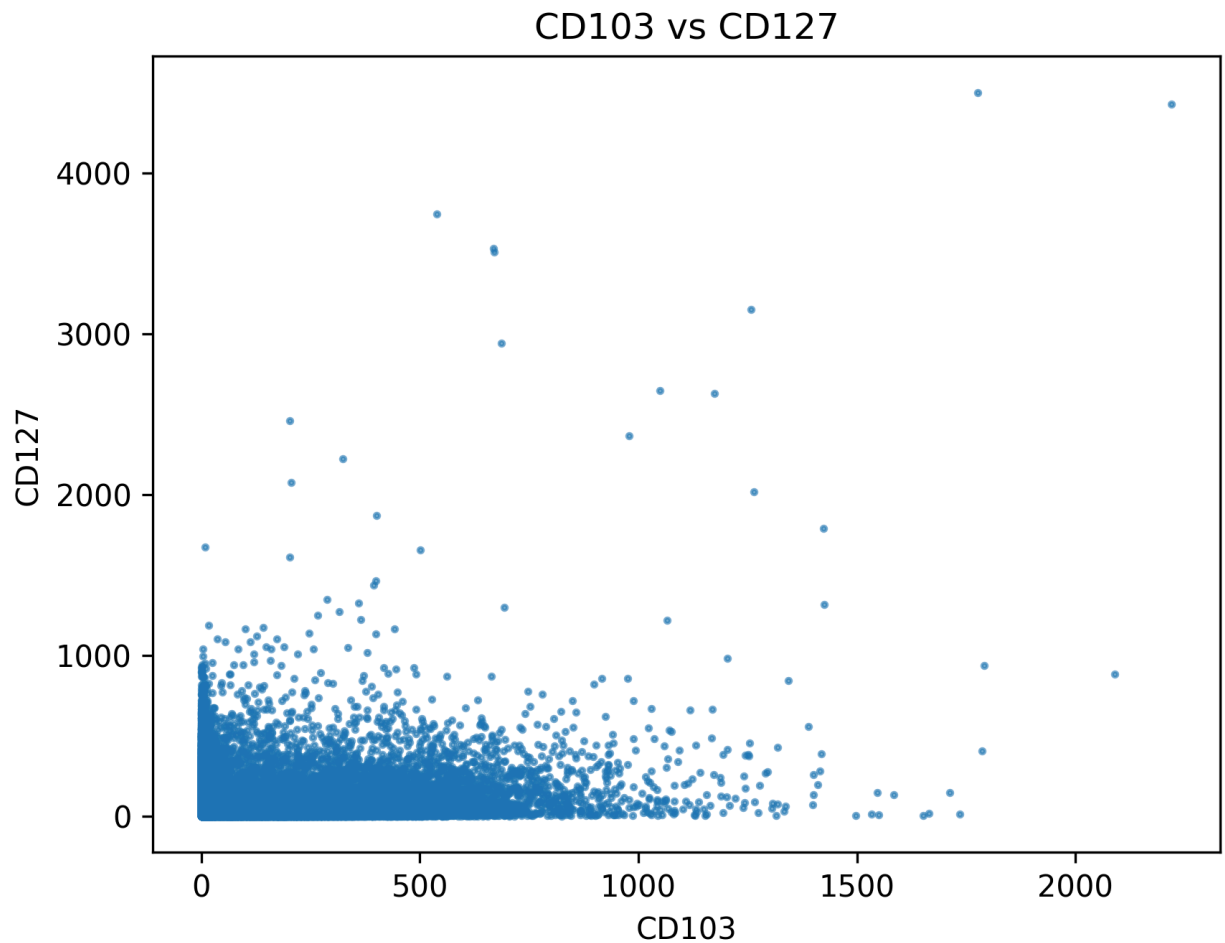
Description: I also looked at the difference in expression for the CD27 gene, which is also a marker. The control and Crohn's disease had similar results, but Crohn's had a bigger IQR. Ulcerative colitis had a smaller IQR than both of them, but its median was similar to both. However, there were many outliers for the ulcerative colitis cells, showing the data is quite centered in that region.

Figure 9: (Alex)



Description: Though it was not a marker, I still decided to explore the difference in expression for the CD28 gene. For all three conditions, the median was at 0. However, the ulcerative colitis differs from the control and Crohn's as it has a higher third quartile, which has around the same amount of expression as the control and Crohn's disease outliers. Though this is not proof of significant difference, it is definitely something to look into.

Figure 10: CD103 Vs CD127 scatter (Eason)

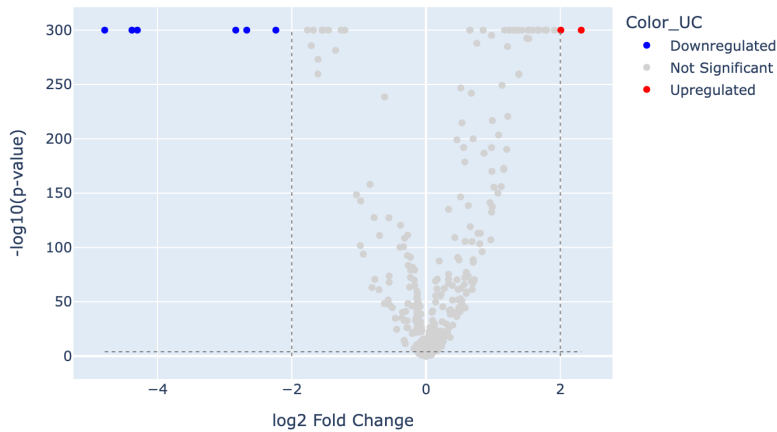


Description: The visualization shows a scatter plot of CD103 and CD127 protein marker expression at the single-cell level. The x-axis represents the measured CD103 expression, and the y-axis represents the measured CD127 expression. Each point represents one cell, and most cells show relatively low expression of both markers.

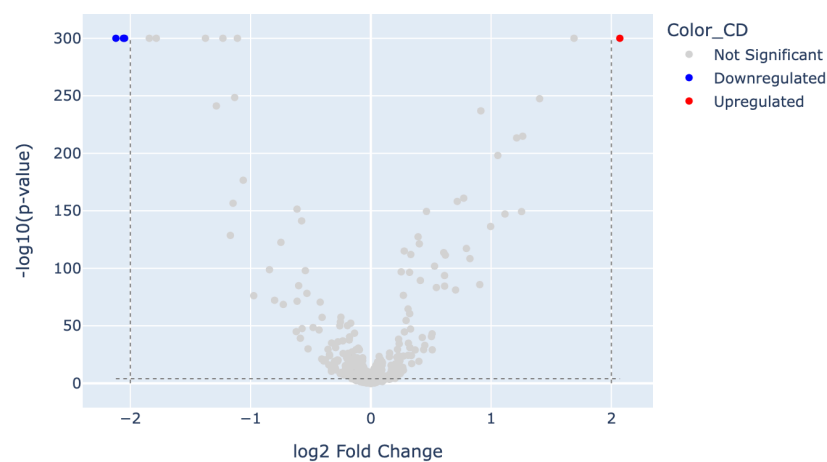
Main finding: The plot suggests that CD103 and CD127 are not strongly correlated linearly across all cells, although some cells show expression in both markers.

Figure 11: Significant Gene Expression Between Disease and Control Volcano Plot (Isabella)

Significant Gene Expression Between Ulcerative Colitis and Control



Significant Gene Expression Between Crohn's Disease and Control



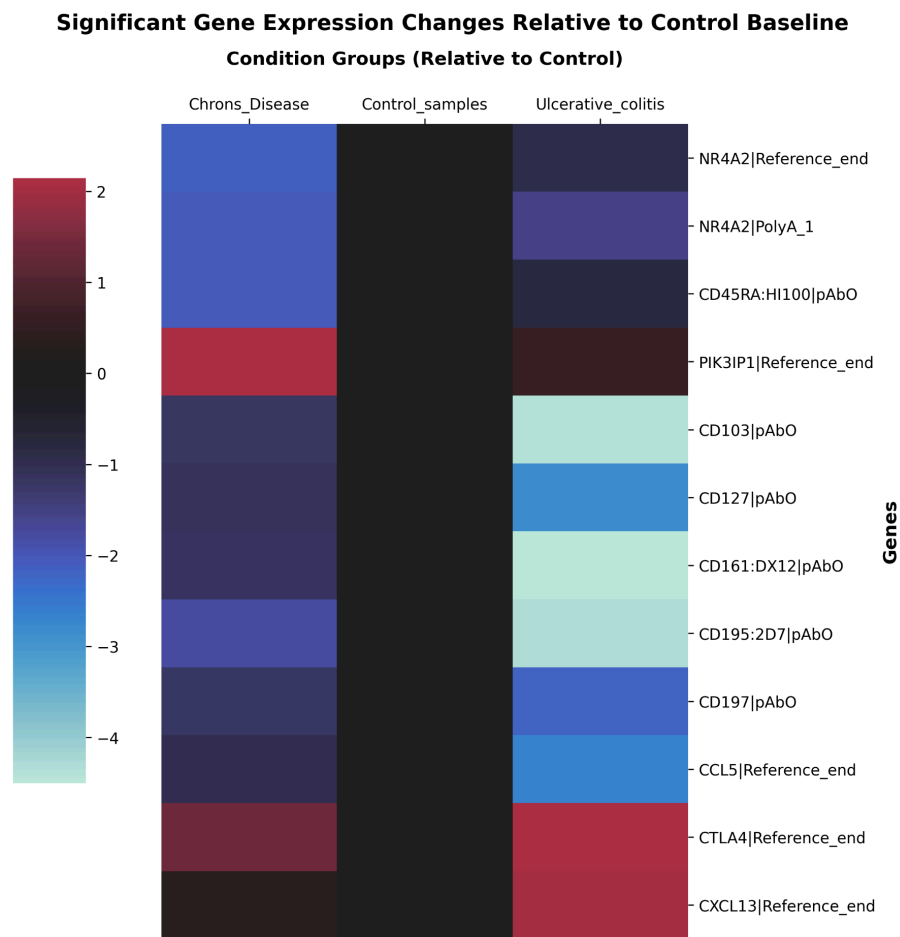
Description:

This is mentioned briefly in the presentation, but it is used to help visualize the thresholds for p-values and log 2 fold transformations. Each dot represents a single gene either in the UC or CD groups. Blue represents a downregulated gene in the disease when it should be more up-regulated in the control. Red represents an upregulated gene that should be more down regulated in the control. The P value threshold is the horizontal line that goes across the bottom of the graph, it is dotted. There are two dotted lines representing the lower and upper bounds for the Log4fold thresholds. Anything to the left of the lower bound and to the right of the upper bound is more than 4 times expressed in the disease in comparison to the control group.

Main finding:

There seems to be a lot of up/down regulated genes with very high p values. This may be because the statistical test used was a Welches' t-test which takes into account the sample size. Because we analyzed 20,000+ cells per group, the p value was very small even for non-significant genes which means much of the analysis for identifying significant DEGs relied on the log 4 fold thresholds. Using this, the genes identified were: NR4A2 (Reference_end and PolyA_1, CD45RA, PIK3IP1 for Chron's Disease and CD103, CD127, CD161, CD195, CD197, CCL5, CTLA4, and CXCL13 for ulcerative colitis.

Figure 12: Significant Gene Expression Changes Relative to Control Baseline (Isabella)



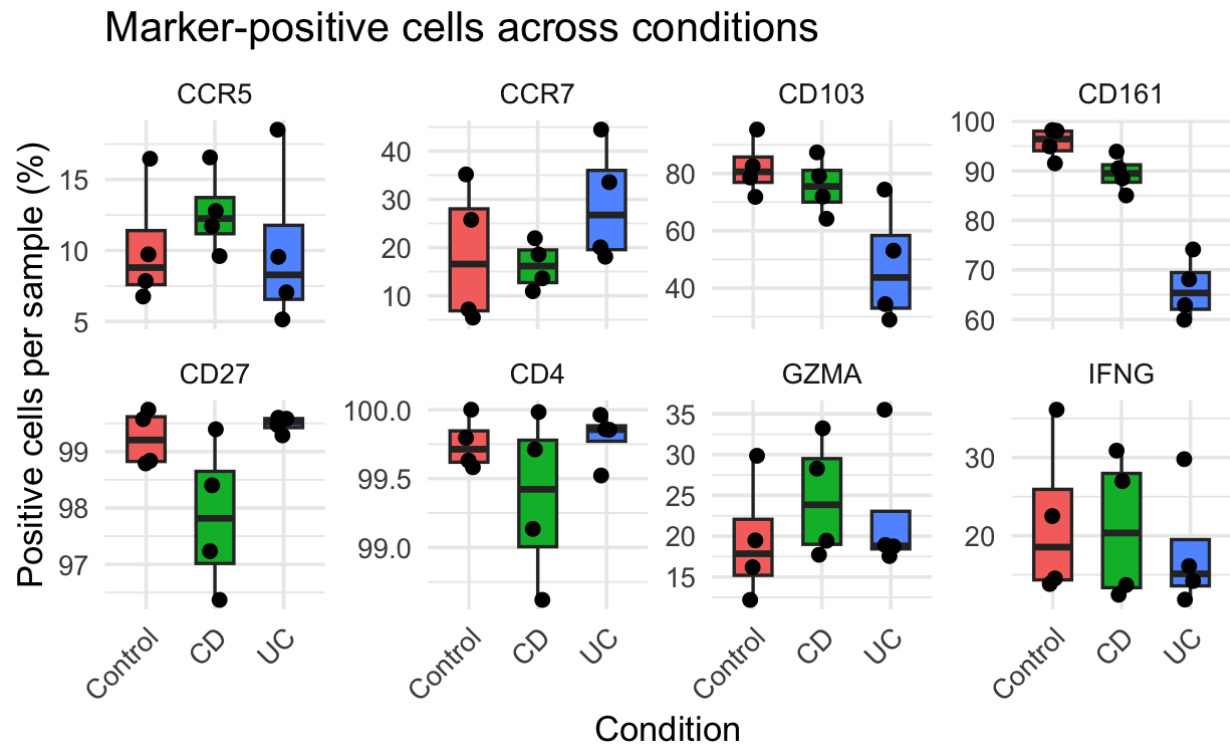
Description:

Each column represented a sample group (Chron’s Disease, Control, Ulcerative Colitis). The genes chosen for the heatmap were obtained via the volcano plot. Each box represents the average gene expression calculated via summing expression for all samples and dividing by sample size. The rows help identify which gene the differences of expression are associated with. The samples closer in color to black have genes that are not differentially expressed. Samples that are red have genes that are more up-regulated. Genes that are a pale blue are extremely down regulated. The legend on the left gives color reference based on the log 4fold transformation.

Main finding:

CD samples showed more significant down-regulated genes compared to up-regulated genes. It is clear that the two diseases likely do not share any genes in common except for CTLA4 which is slightly upregulated in Chron’s disease and significantly more upregulated in ulcerative colitis. It appears the CD

Figure 13: Marker-positive cells across conditions (Jiayan)

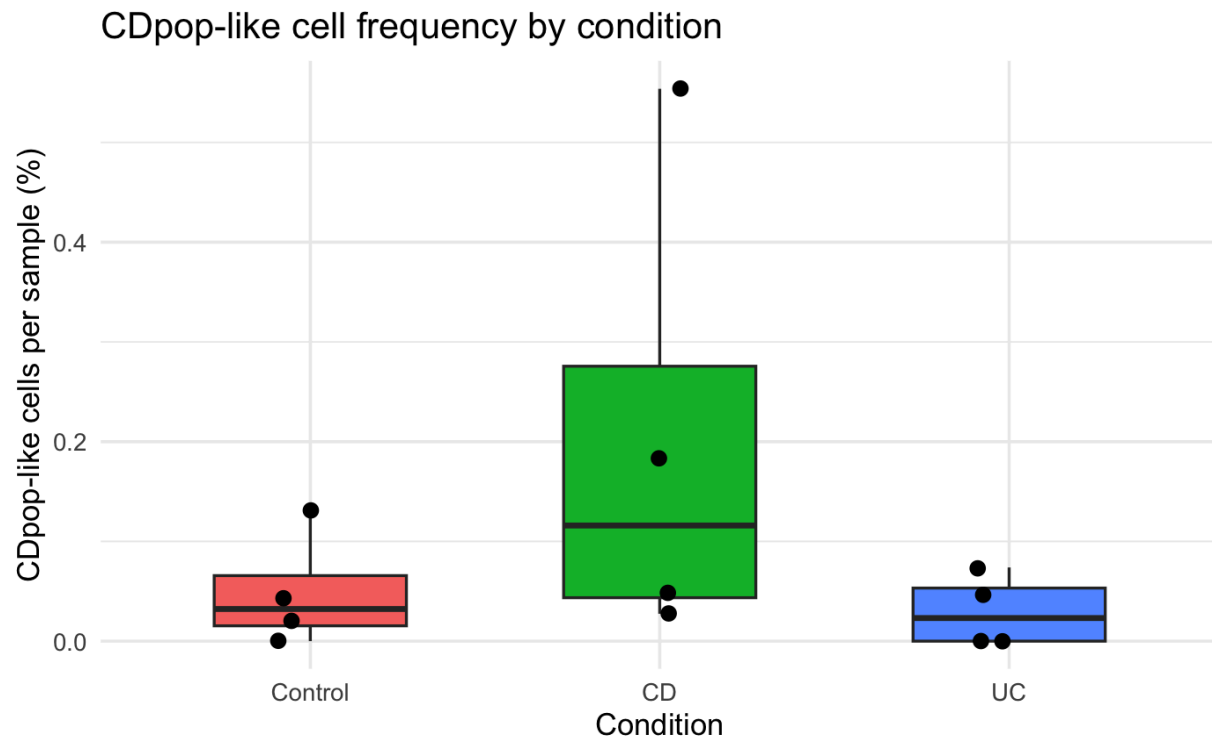


Description: This figure shows the percentages of cells positive for each selected marker across CD, UC and Control samples. Each dot represents one patient sample, and the boxplots summarise the sample-level distribution within each condition. Positivity was defined as marker signal greater than 0.

Main findings:

- CD103 and CD161 positivity appeared lower in UC samples compared with Control and CD
- CCR7 positivity appeared higher in UC samples
- GZMA and IFNG showed variation across samples, suggesting functional marker activity differs between individuals
- CD4 and CD27 were detected in most cells across all conditions
- After pairwise Wilcoxon tests with Benjamini-Hochberg correction, none of the marker differences were statistically significant, so these patterns should be interpreted as descriptive trends

Figure 14: CDpop-like cell frequency by condition (Jiayan)



Description:

Each point represents the percentage of cells classified as CDpop-like within one patient sample. CDpop-like cells were defined using the simplified marker pattern CD4+ CD103+ CD161+ CCR5+ CD27- CCR7-. The simplified detected/not-detected threshold considered CD4, CD103, CD161, and CCR5 to be positive if their signal was greater than 0, while CD27 and CCR7 were considered negative if their signal equaled 0. The boxplots summarize the four samples in each condition.

Main finding:

Observed a higher frequency of CDpop-like cells in CD samples than Control and UC samples, but there was substantial variation among CD samples.